

\$4401B

Passive S&P 500 AUM (latest)

30

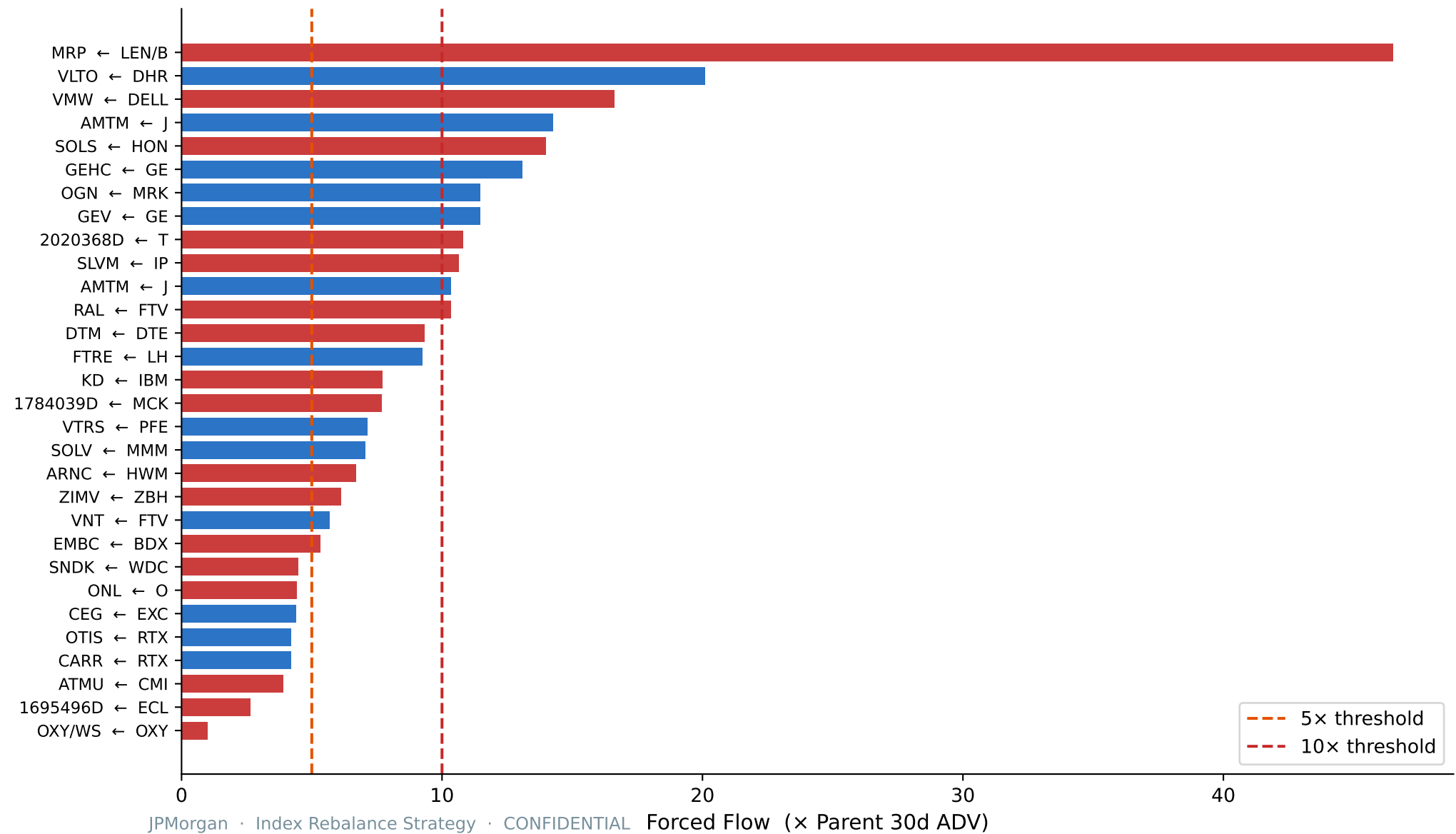
Spinoff events (2020-2024)

17 / 30

Children excluded from S&P 500

7.7x

Median forced flow / parent ADV



excluded

immediately included

later included

Passive AUM × Parent Wt

Parent 30d ADV

FF % Float:

Forced Flow (\$)

Child Mktcap at t=0

Gate: child must be EXCLUDED from S&P 500 (passive must sell)

n=17 excl · n=13 imm · n=0 later

+1.8%

Excluded mean short P&L t=21d (n=9)

-4.0%

Excluded mean short P&L t=42d (n=9)

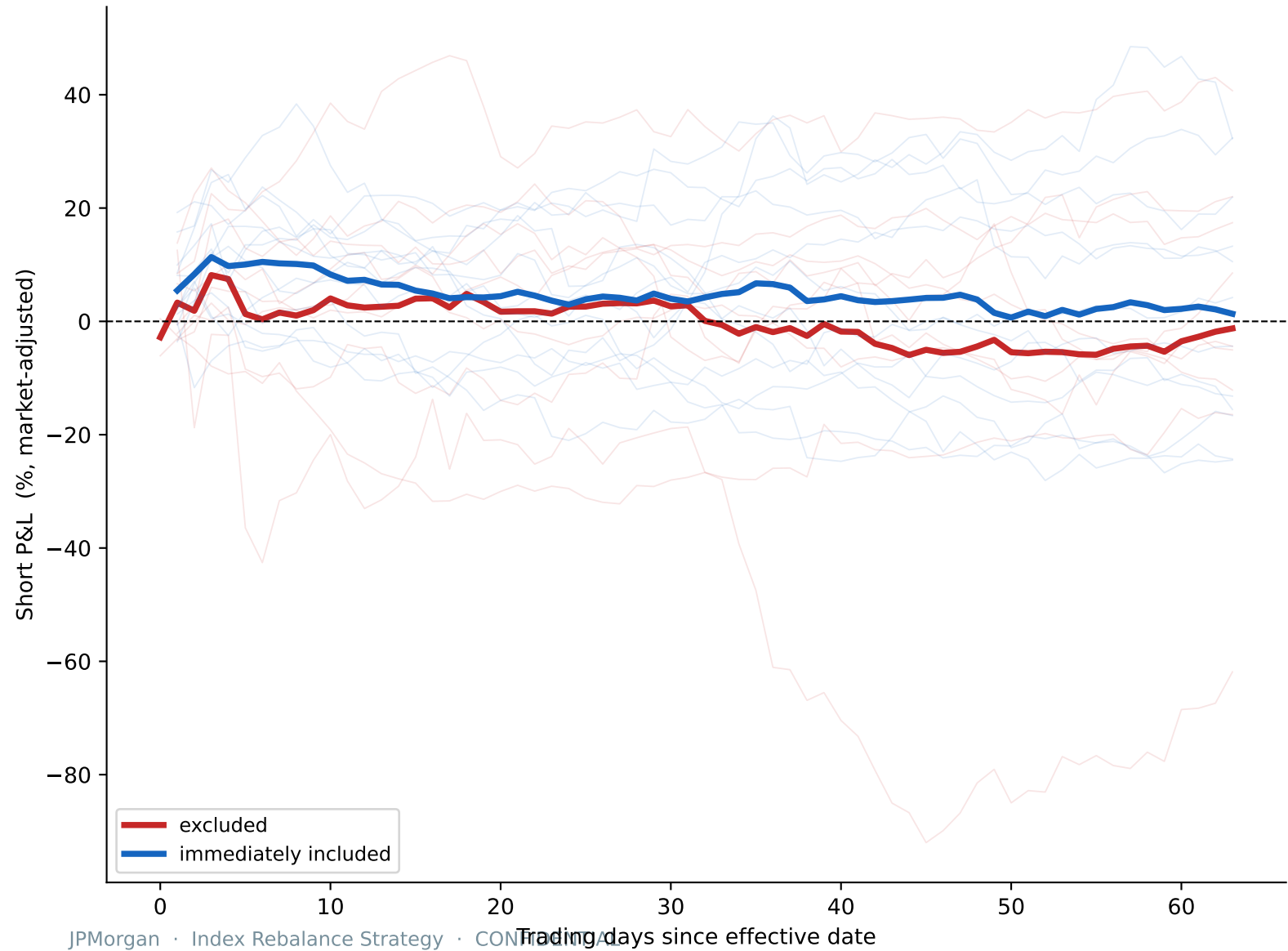
56%

Win rate t=42d (excluded)

p=0.725

t-test vs zero t=42d

Short P&L by Child Inclusion Status



Mean Short P&L by Horizon

Horizon	Excl.	Incl.	p-val
t=5d	+1.3%	+10.8%	0.832
t=10d	+4.0%	+8.5%	0.537
t=21d	+1.8%	+5.0%	0.795
t=42d	-4.0%	+5.3%	0.725
t=63d	-1.2%	+3.7%	0.901

+0.21

FF×ADV vs t=42d return correlation (excl.)

-4.2%

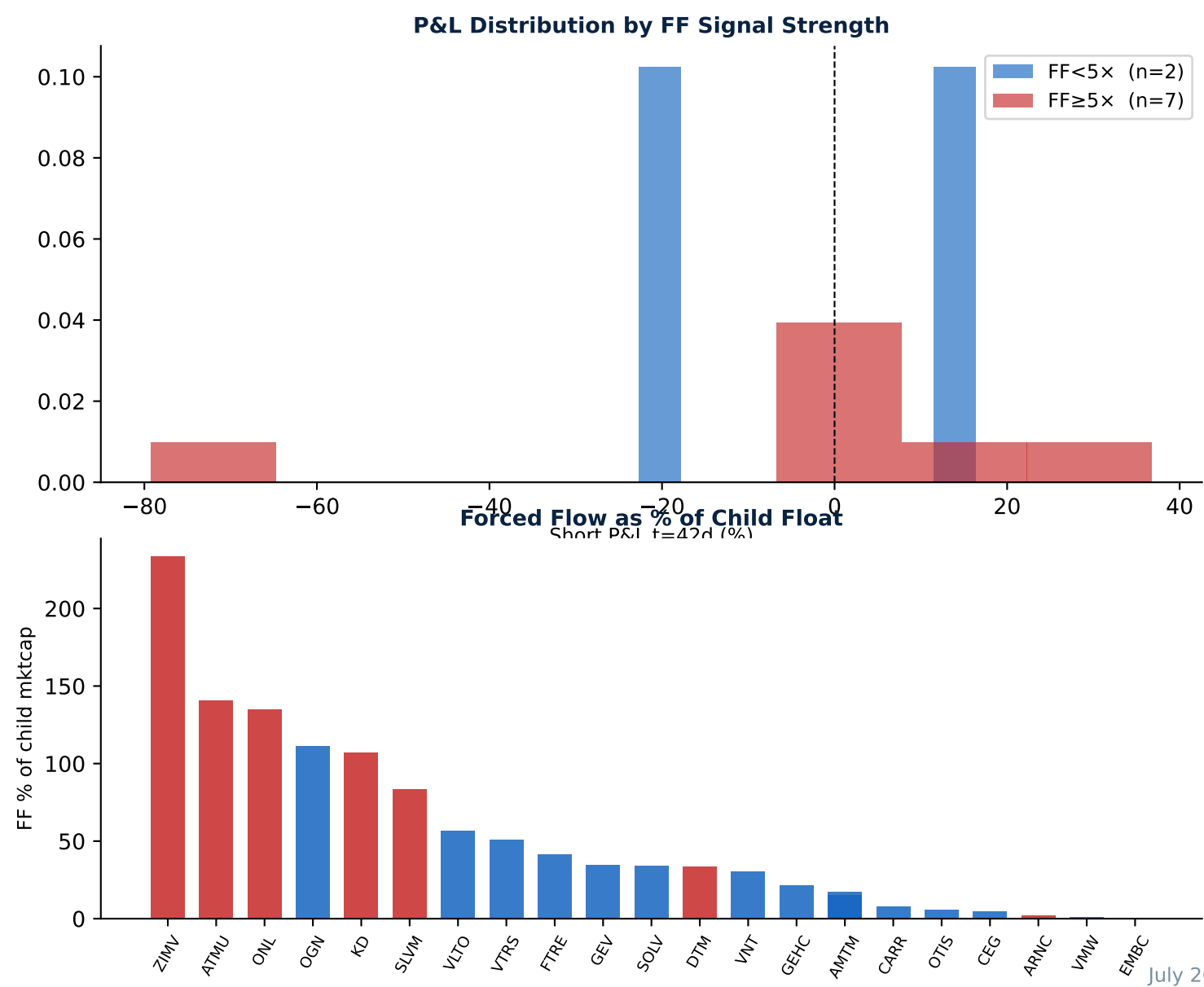
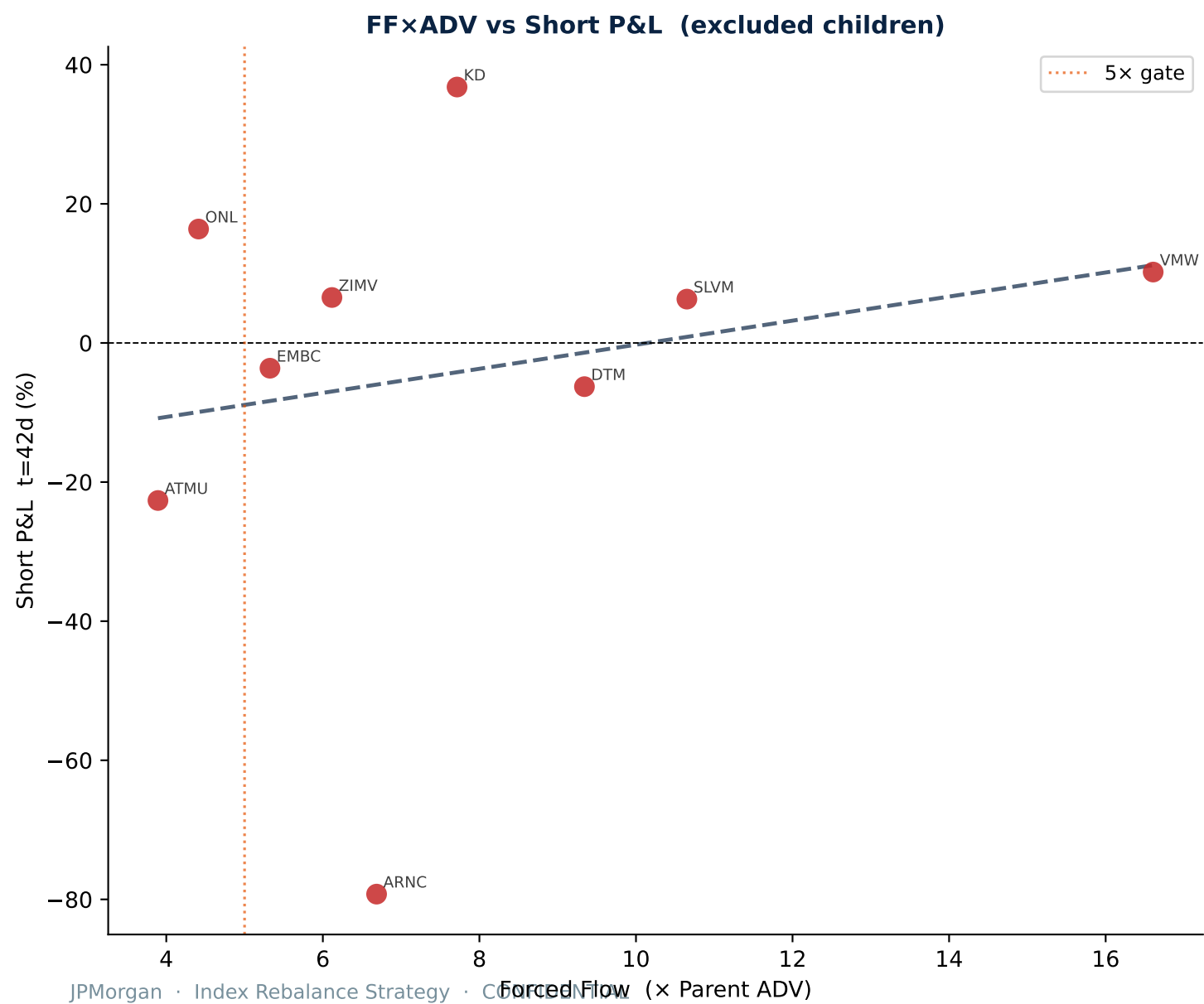
Mean P&L FF≥5× (n=7)

-3.1%

Mean P&L FF<5× (n=2)

83.4%

Median FF% of child mktcap (excl.)



-0.2%

Avg parent ann.-day CAR (n=28)

-2.7%

Avg parent CAR announce → effective

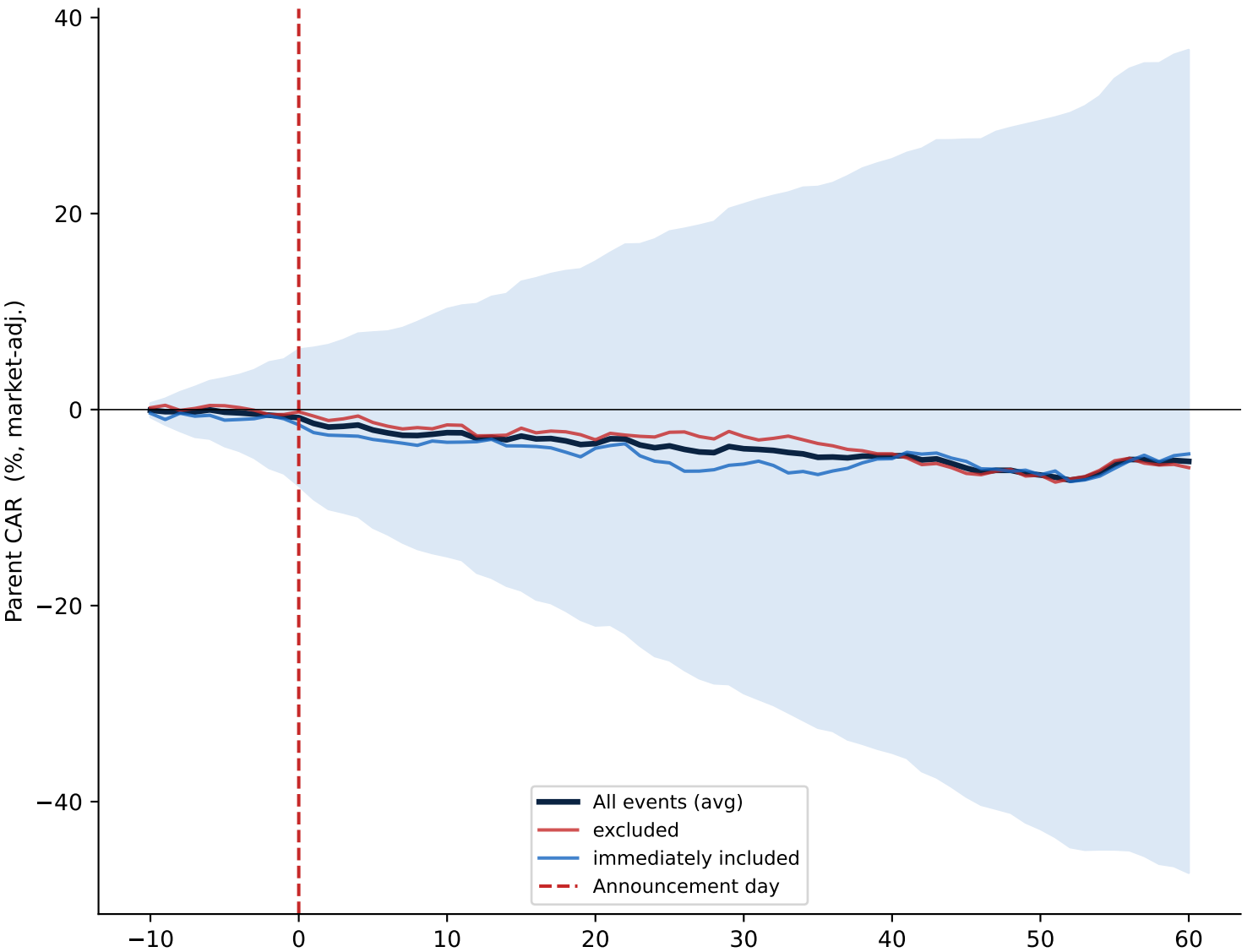
29%

Win rate announce → effective

338d

Median announce → effective lag

Parent Stock: Announcement-Centered CAR



Parent	Child	Lag	Ann Day	Ann→Eff	Status
WDC	SNDK	483d	+6.1%	+24.2%	excl
HON	SOLS	387d	+0.8%	+8.3%	excl
ECL	1695496D	486d	-0.5%	+6.4%	excl
J	AMTM	312d	-0.5%	+5.8%	imm
PFE	VTRS	141d	+0.4%	+5.6%	imm
IP	SLVM	302d	-3.2%	+4.4%	excl
MRK	OGN	484d	-4.0%	+1.9%	imm
ZBH	ZIMV	389d	-0.9%	+0.4%	excl
RTX	CARR	494d	-2.5%	-1.0%	imm
RTX	OTIS	494d	-2.5%	-1.0%	imm
O	ONL	171d	+0.5%	-1.6%	excl
J	AMTM	16d	-0.5%	-2.0%	imm
BDX	EMBC	330d	-0.9%	-2.2%	excl

-4.0%

SPY hedge P&L t=42d (n=9)

-0.7%

Pairs P&L t=42d (n=9)

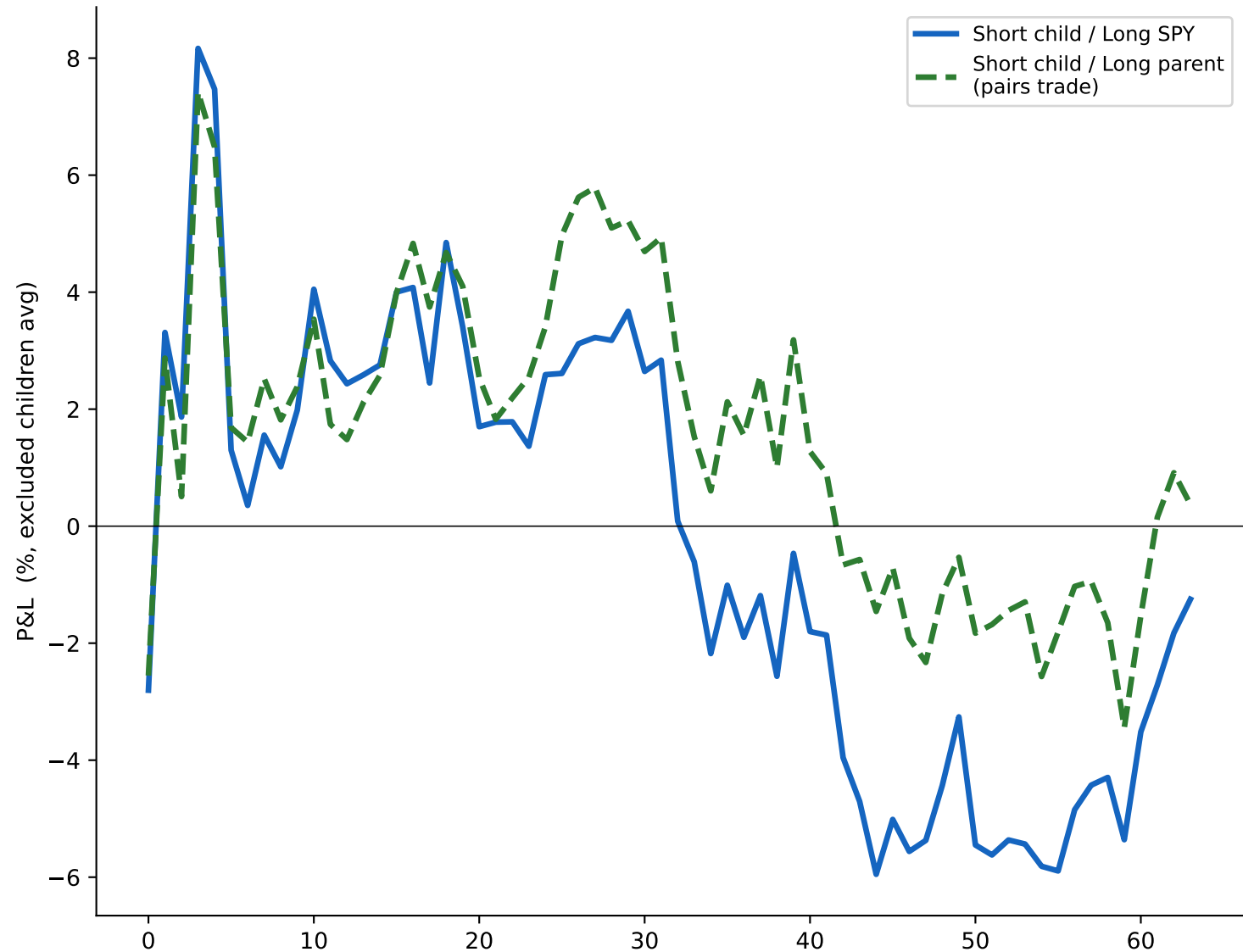
56%

Pairs win rate t=42d

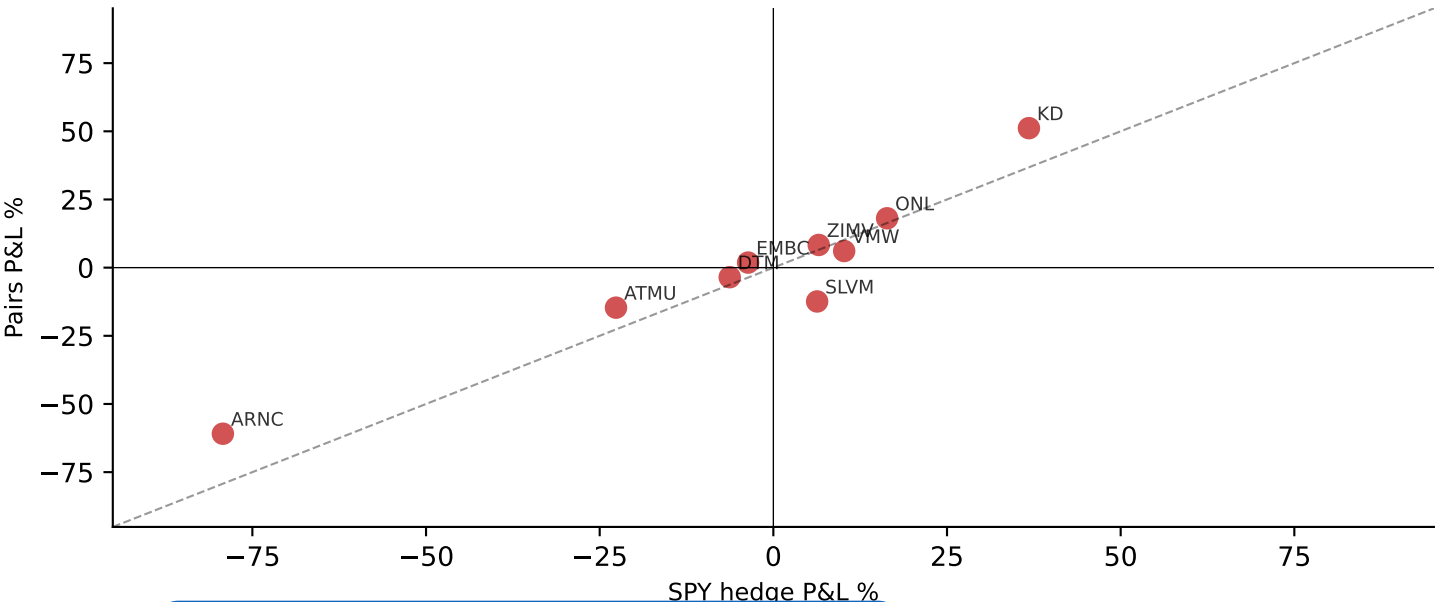
+3.3%

Pairs vs SPY hedge differential

SPY Hedge vs Pairs Trade — Avg P&L Path



SPY vs Pairs (t=42d, per event)



Pairs rationale:

- Parent & child share sector/macro exposure
- Long parent hedges both SPY beta and sector
- Residual P&L $\approx$ pure forced-selling alpha
- Exit: child S&P 500 addition OR 42-day stop

0.84

Median child β vs S&P 500 (OLS t=1-60)

0.94

Mean child β

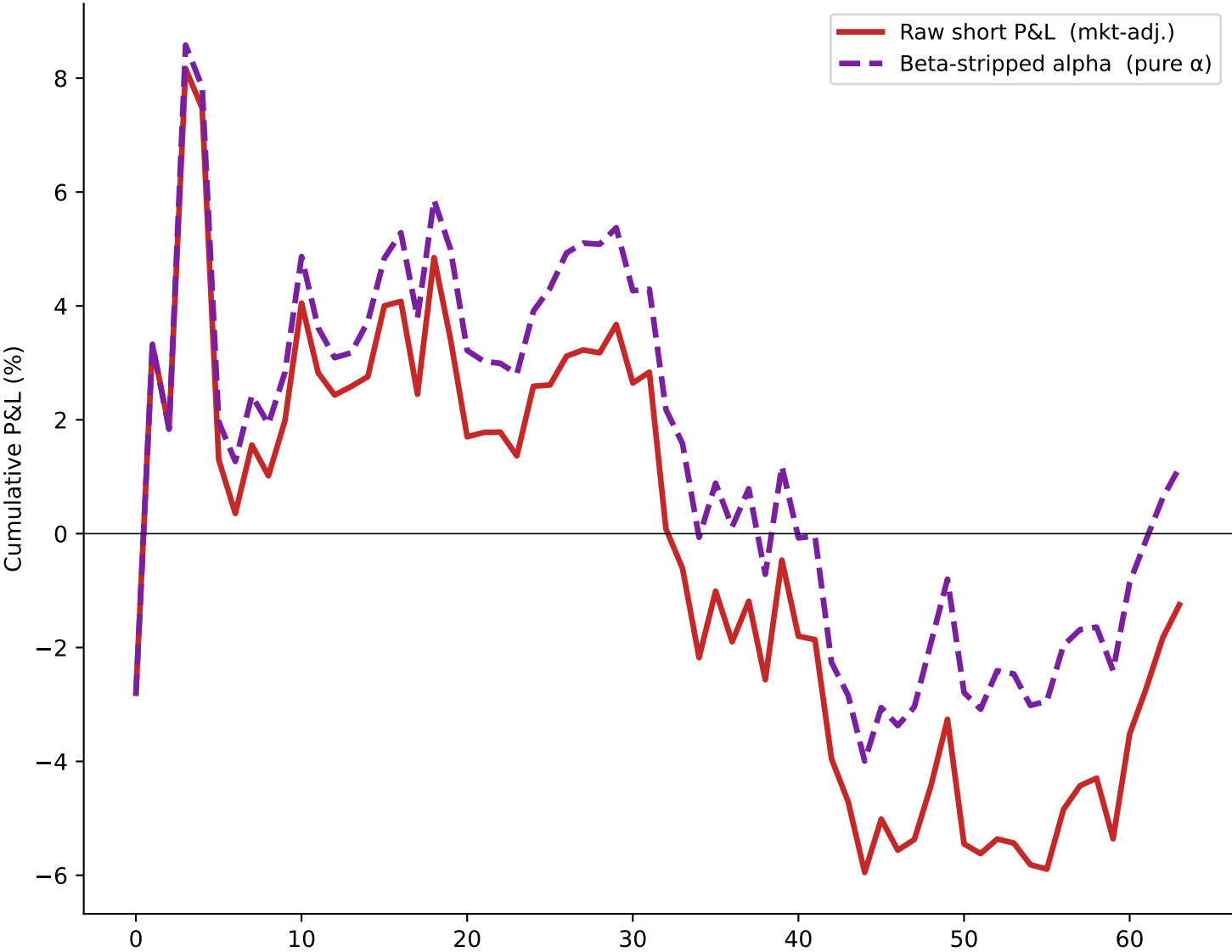
Alpha-driven

Primary source of underperformance

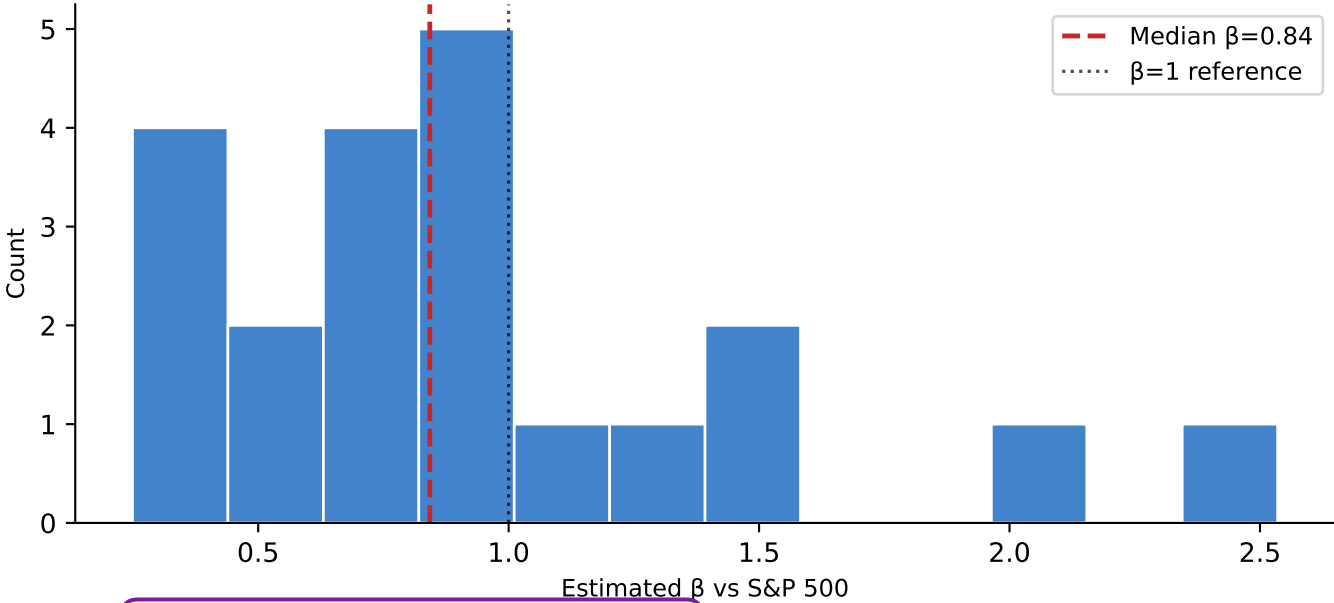
21

Events with β estimate (≥ 10 obs)

Raw P&L vs Beta-Stripped Alpha
(excluded children, equal-weighted avg)



Child β Distribution



Beta estimation:

- OLS: child ret = $\alpha + \beta \times \text{S\&P 500 ret}$
- Window: t=1 to t=60 post-effective
- $\beta < 1$ for most excluded children (forced selling depresses price, not index co-movement)
- Live trading: use parent pre-spinoff β to avoid ex-post lookahead

71%

LOO-CV accuracy (n=21)

0.760

ROC AUC

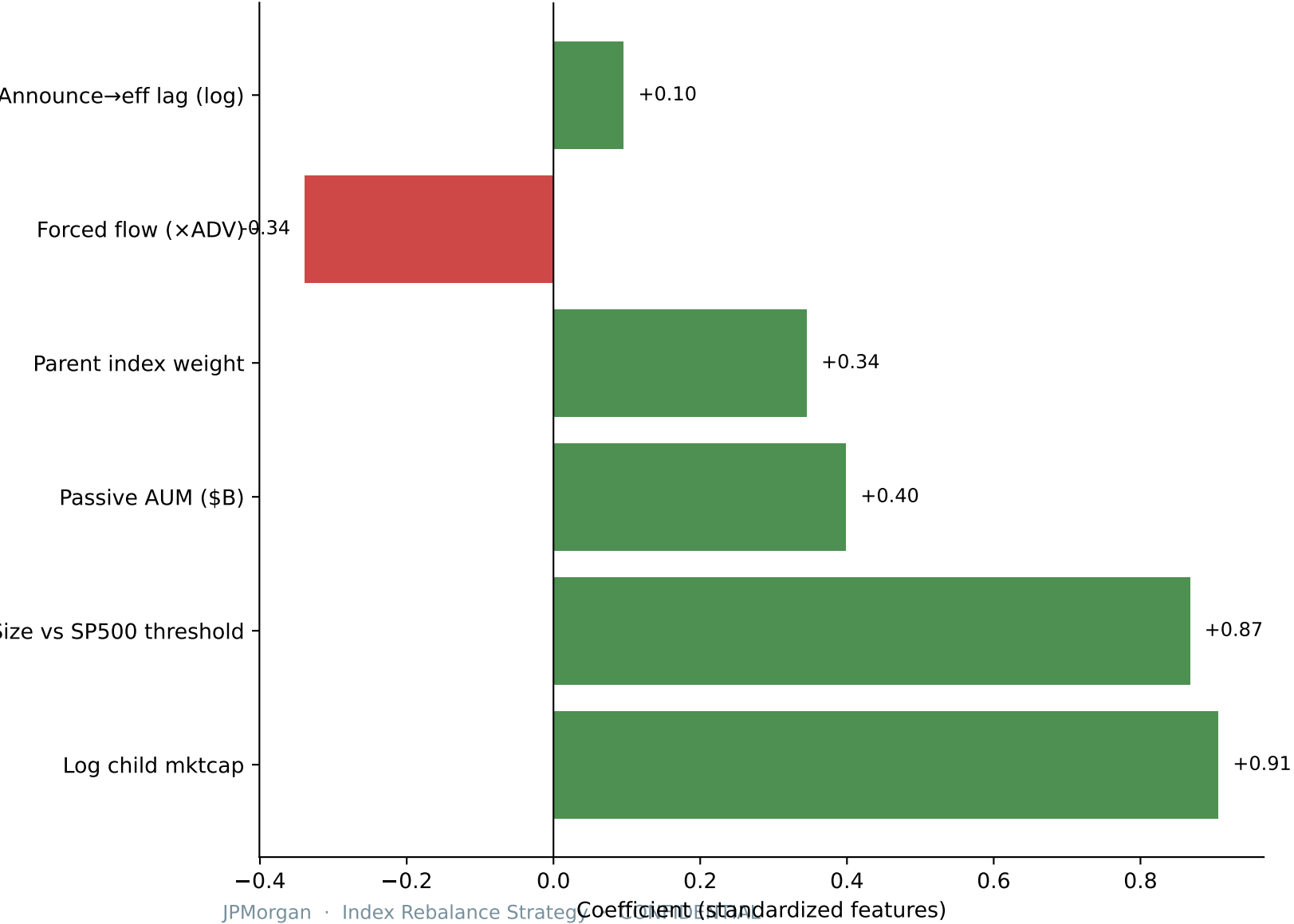
38%

Baseline accuracy (always predict excl.)

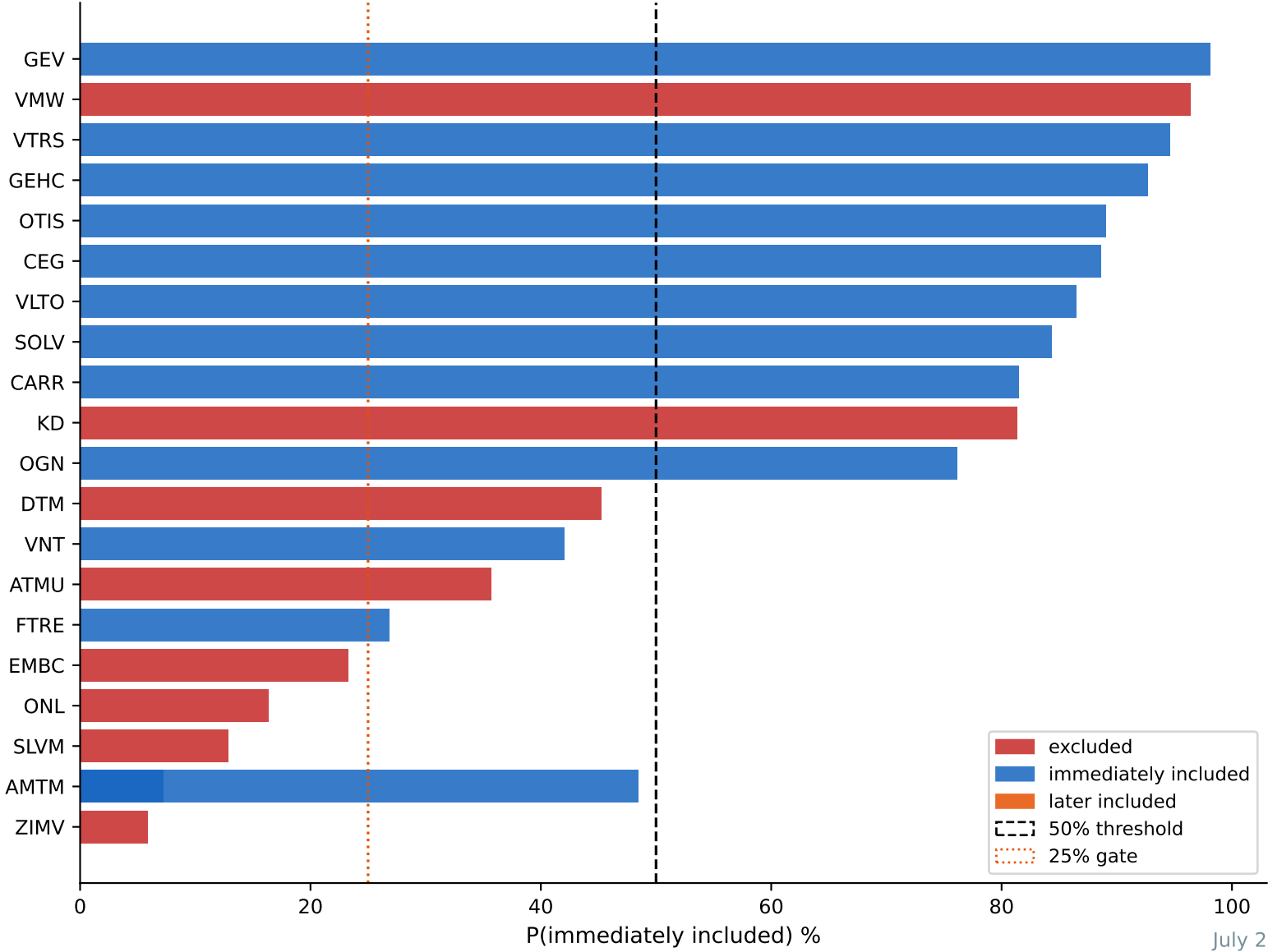
13

Immediately-included events in sample

Feature Importance
(+ → predicts immediate inclusion)



LOO-CV Predicted Inclusion Probabilities



-4.0%

S1 (All excl.) mean P&L (n=9)

56%

S1 win rate t=42d

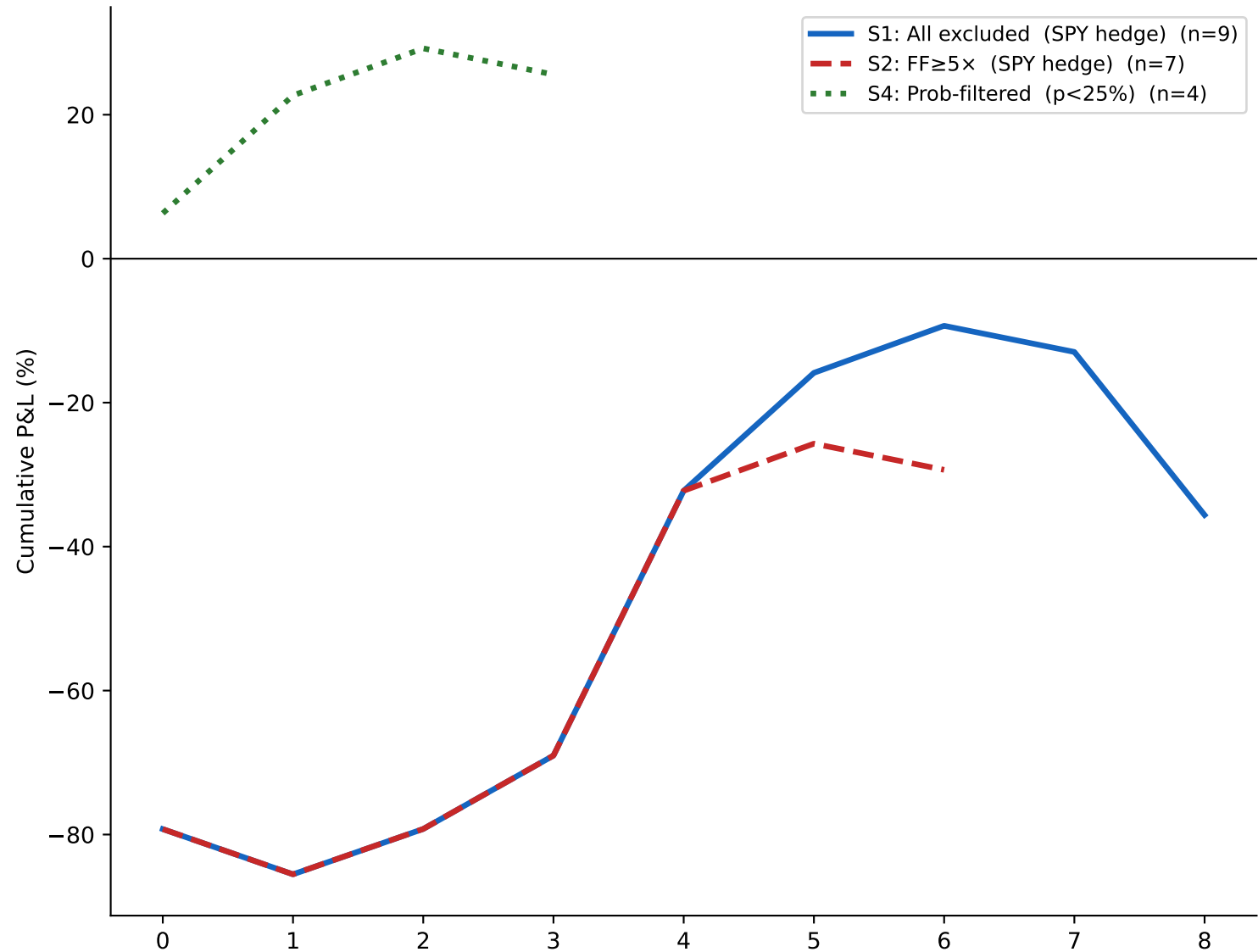
-4.2%

S2 (FF≥5×) mean P&L (n=7)

57%

S2 win rate t=42d

Equity Curves — 42-Day Time Stop



Strategy	N	Mean P&L	Win%	t-stat	p-val
S1: All excl.	9	-4.0%	56%	-0.36	0.725
S2: FF≥5×	7	-4.2%	57%	-0.31	0.768
S4: p<25%	4	+6.4%	75%	1.57	0.215

Signal Priority

- 1. Gate: inclusion_status = excluded (passive must sell)
- 2. Sizing: FF% float (forced flow ÷ child mktcap)
- 3. Signal: forced_flow_adv (× parent 30d ADV)
- 4. Filter: prob_included < 25% (logistic model gate)
- 5. Hedge: long parent >> long SPY (pairs trade)